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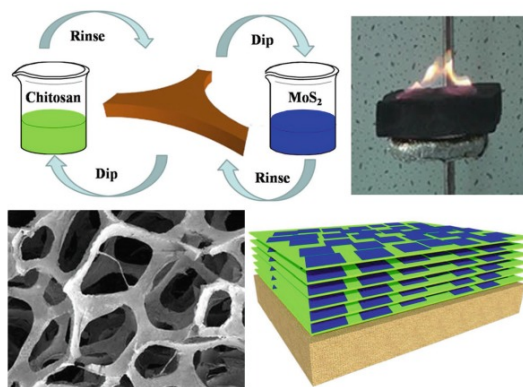
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Review



Oxide-based RRAM materials for neuromorphic computing

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ABSTRACT

In this review, a comprehensive survey of different oxide-based resistive random-access memories (RRAMs) for neuromorphic computing is provided. We begin with the history of RRAM development, physical mechanism of conduction, fundamental of neuromorphic computing, followed by a review of a variety of RRAM oxide materials (PCMO, HfO_x, TaO_x, TiO_x, NiO_x, etc.) with a focus on their application for neuromorphic computing. Our goal is to give a broad review of oxide-based RRAM materials that can be adapted to neuromorphic computing and to help further ongoing research in the field.

Introduction

Resistive random-access memory (RRAM) utilizes the resistive switching (RS) phenomena to store information, which offers new types of devices that can outstrip the performance of traditional semiconductor electronics devices. Compared to charge-based memory devices, RRAM stands out due to its smaller cell size $4F^2$, multi-bit capability as well as energy per bit ($\sim$ fJ/bit). The first report of resistive switching phenomena is by Hickmott [1], where resistive switching is found on SiO_x, Al₂O₃, Ta₂O₅, ZrO₂ and TiO₂. However, it was only after 38 years that the heat of RRAM research was reawakened with the observation of resistive switching behavior in magnetoresistive films from University of Houston

[2]. Two years later, 64-bit RRAM array using Pr_{0.7}Ca_{0.3}MnO₃ via a 500-nm complementary metal oxide semiconductor (CMOS) process was reported by Zhuang [3]. Between 2004 and 2007, Samsung and Infineon [4] made a significant progress on RRAM development with the first 3D RRAM array demonstrated in 2007. The concept of RRAM used for neural network and logic circuit was first published in *Nature* by HP in 2008. The paper titled “The Missing Memristor Found” [5] triggered another heat of RRAM development. In the next 9 years, great achievement had been witnessed from industries and academies. The successful fabrication of 64-MB RRAM test chip, 32-Gbit bilayer cross-point RRAM, 27-nm 16-Gbit Cu-based CBRAM test chip and 4-layer 3D vertical self-selective RRAM array was

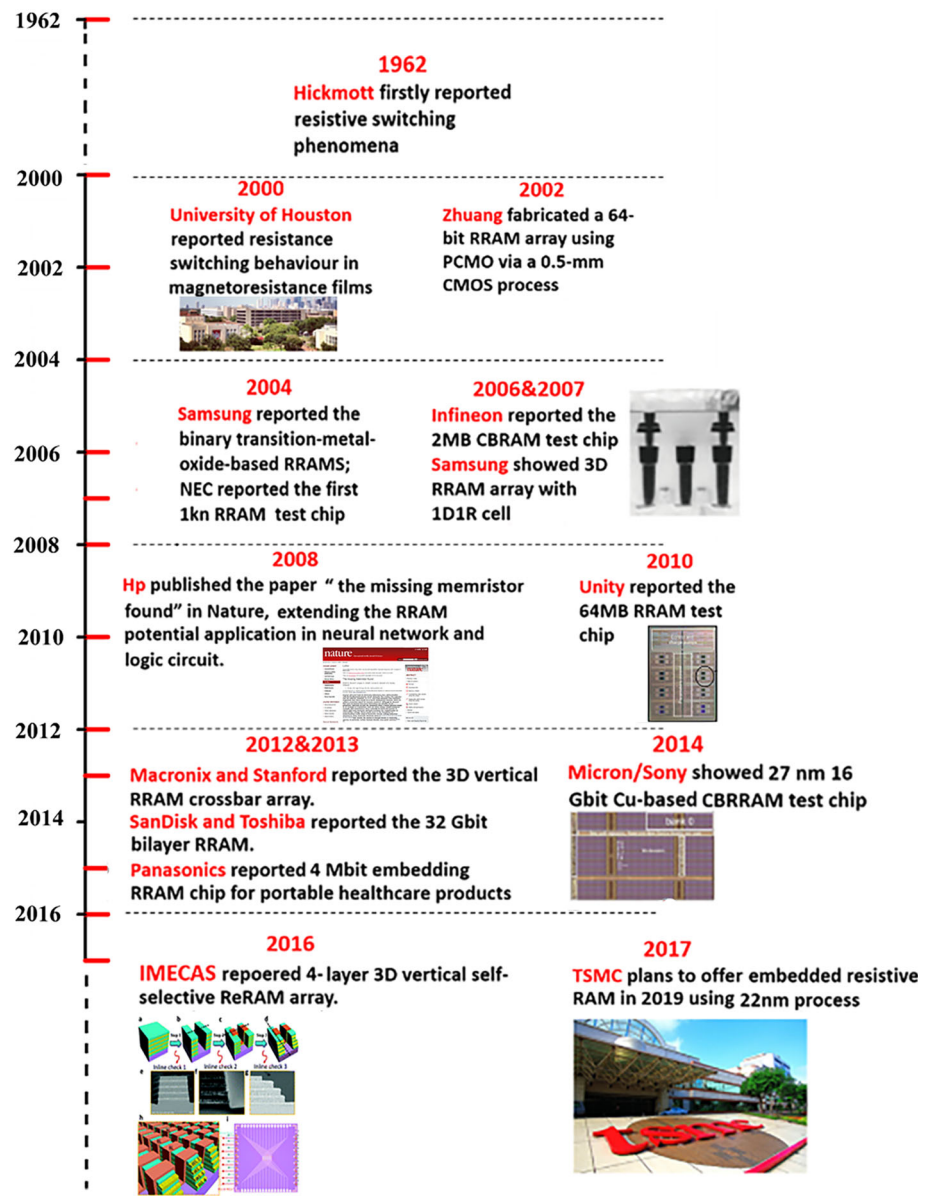
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announced by Unity [6], SanDisk/Toshiba [7], Micron/Sony [8] and IMECAS [9], respectively. In 2017, TSMC [10] announced its new plan to start producing embedded RRAM chips in 2019 using a 22-nm process. Figure 1 shows the historical timeline of important events on RRAM development in the last half-century.

RRAM technology is compatible to the conventional CMOS in a simple way [11], and the RRAM embedded system could be integrated into IoTs, automobile and infotainment platforms [12, 13]. In addition to the memory hierarchy, RRAM has demonstrated its application in the low-power computing as non-volatile logic circuits and

neuromorphic computing as a synaptic device. Recent reviews by H.S.P. Wong, Yu, S. M. and Akinaga, H [14–16] gave excellent and comprehensive overviews of RRAM physical mechanism, materials, performance and applications. Here, we limit this review to the oxide-based RRAM materials where neuromorphic computing applications have been demonstrated. At first, fundamentals of RRAM and neuromorphic computing will be discussed. Thereafter, we will elaborate more on these oxide-based RRAM devices (PCMO, HfO_x , TaO_x , TiO_x , etc.) used for neuromorphic computing application. This review will end with a conclusion and outlook.

Figure 1 Historical timeline of RRAM development from 1962 to 2017 [1–10].



RRAM design and physical mechanism

A typical RRAM structure consists of three layers of materials, namely (metal–insulator–metal) MIM structure [17], which is a resistive oxide layer sandwiched between two electrodes. The resistance of the device can be modulated by applying an external electric field across the electrodes. RRAM devices work based on chemical redox reactions, i.e., oxidation and reduction reactions. In memory storage, there are binary states, “0” and “1”. “0” represents data that is not stored, while “1” represents data that is stored. In RRAMs, redox reactions form a filament or bridge across the two metal layers, within the insulator. When a filament is created, an electrically conductive is formed between the two metal layers. Hence, a low-resistance state (LRS) occurs when a filament is formed, while a high-resistance state (HRS) occurs when a filament is not formed or ruptured. When a filament is formed, it would be a “1” state where one bit of information is stored and when a filament is ruptured, it would be a “0” state [18] (Fig. 2). A typical hysteretic current–voltage (I – V) characteristics in a metal–insulator–metal (MIM) structure are shown in Fig. 3. The switching from HRS to LRS is called the SET process, while the switching from LRS to HRS is called RESET process. In most situations, a forming voltage larger than the SET voltage is needed for the pristine device to trigger the resistive switching behaviors for the subsequent cycles. The SET and RESET processes can occur at the same polarity of the applied voltage as shown in Fig. 3a which is known as unipolar switching. Similarly, bipolar switching refers to the switching

behaviors occurring at different polarities of voltage (Fig. 3b).

The MIM stack of RRAM can be broadly classified as symmetric and asymmetric structures. A symmetric MIM structure mainly exhibits unipolar switching behavior, while a asymmetric structure mainly exhibits bipolar switching behavior. The mechanism of resistive switching behavior can mainly be classified into the electrochemical metalization effect (ECM) and the valence change memory effect (VCM) [19]. ECM and VCM are typically observed in RRAMs with one active metal electrode (Cu, Ag, Ni) and one inert metal electrode (Pt, Ru, Au or Ir). In ECM, the conductive path of the switching layer is formed via metal cations of the electrochemically active metal electrode under an external electric field. On the other hand, VCM works based on the migration of anions where oxygen vacancies contribute to the conductive path within the oxide layer. In VCM, it typically requires an oxygen scavenging electrode to facilitate anionic movement between the active and inert electrodes. ECM and VCM are based on redox reactions of oxidation and reduction. RRAM devices with switching layer doped with Ti [20] or Ge [21] had been reported to have a forming-free property where their initial resistance state (IRS) is similar to the high-resistance state (HRS). In addition, several other advantages of doping in the RRAM fabrication process to improve uniformity and frustration had been reported [22]. Besides the single switching layer, numerous investigations had been conducted on double and multiple switching layers. It has been reported that a switching layer consisting of low-resistance and high-resistance layers can reduce the randomness of resistive switching. Additionally, multi-layer structures have been promising for the multi-level storage function [23, 24]. Table 1 summarizes the details of the RRAM stack, electrodes and switching layers.

The conduction mechanism in oxide RRAM can be analyzed by fitting the I – V characteristic of current conduction in the HRS and the LRS. Various resistive switching mechanisms have been proposed to explain and model RRAM conduction behavior. They include the Pool–Frenkel emission (P–F emission), SCLC (spaced charge limited current), formation and rupture of conductive filaments (CFs), electrode-limited conduction (Table 2) [42]. Although the underlying physical mechanisms are material/electrode specific and not well understood, the

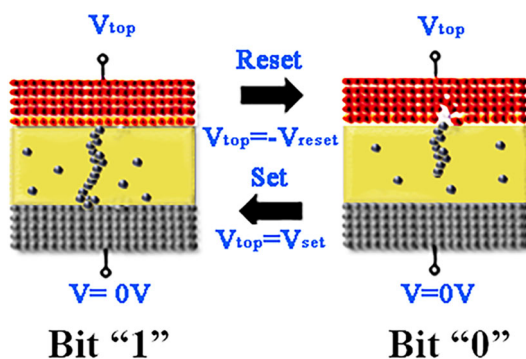
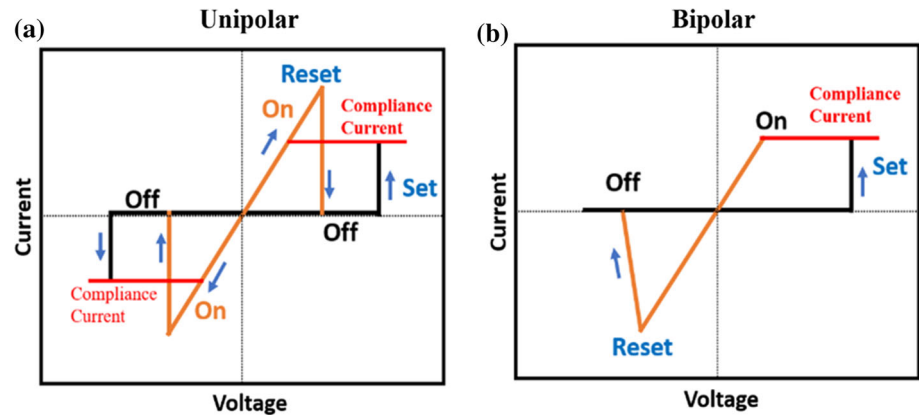


Figure 2 A schematic diagram of the mechanism of resistive switching effect in RRAM cell, “1” state when a filament is formed and “0” state when a filament is ruptured.

Figure 3 Typical hysteretic current–voltage (I – V) characteristics in metal–insulator–metal (MIM) structures **a** unipolar switching and **b** bipolar switching.



conductive filament mechanism applies to a large majority of binary oxide unipolar or bipolar RRAMs.

Neuromorphic computing

Comparing with traditional computational architecture, the human brain working system has major advantages in the following aspects. The energy efficiency of the human brain (10 watts) [15] is remarkably superior, reaching 5 orders of magnitude lower than supercomputers. The human brain can also adapt to different environments and is able to perform complex processing. The advancement of studies on the human brain system has led to a new disruptive technology neuromorphic computing. This technology imitates how the brain is working in the design and implementation of the computational system.

The ultrahigh energy efficiency of the human brain is derived from its low-power neurons and spike-based computation, which is mainly realized by four components (Fig. 4). Neuron bodies function as an integrator and a device for threshold spiking. In order to implement a neuromorphic system, a capacitor is used to mimic neuron bodies in neuromorphic architecture. Axons work as information transmission connection which could be mimicked a long wire. The signal input from multiple neurons to a single neuron is provided by dendrite, a short wire which plays the role of dendrite in the neuromorphic system. The synapse is the most investigated component that has been constructed so far, and it provides dynamical interconnections between neurons by switching and plasticity. Memristor is the device component that works as a substitution to a synapse.

In the neuroscience, synaptic weight of synapse refers to the strength or amplitude of a connection between two neurons which is mainly determined by the amount of neurotransmitter released and absorbed. The synaptic plasticity is the increase or decrease of synaptic weight over the time. Short-term synaptic plasticity acts on a timescale of tens of milliseconds to a few minutes, while long-term plasticity lasts from minutes to hours. Memristors are electrical devices which can mimic the synapse. The resistance of memristors, which functions as the synaptic weight in the neuroscience, can be tuned as response to a periodic voltage (or current) input, and the resistance state can be retained when the power is turned off. RRAM is one form of memristor, and its resistance (synaptic weight) can be manipulated between high and low resistive states by deliberately applied voltage, which causes the medium to acquire microscopic conductive paths called filament. In addition, it is possible to reproduce synaptic plasticity with RRAM memory run by specific coding schemes. In the human brain, 100 billion neurons are connected to one another by 100 trillion synapses [43]. The neuromorphic computing processor consists of a vast array of electronic spiking neurons and multi-bit synapses realized using fully interconnected crossbar memories.

There is a very large amount of fascinating work performed to develop memristors for neuromorphic systems. So far, several types of memristors have been investigated including but not limited to RRAM [44–49], phase change memory [50–61], spin device [62–68], floating gate transistors [69–76] and an optical device [77–81]. Among them, RRAM is an excellent memristor for non-volatile synapse application as it enables higher density, scalability and efficient

Table 1 Summary of RRAM stack, electrodes and switching layer

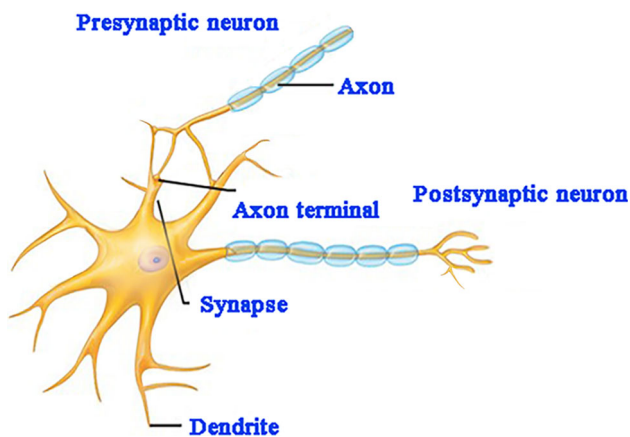
	Structures	Switching behavior	Model
<i>RRAM stack</i>			
Symmetric and asymmetric	Symmetric structure: (e.g., Pt/NiO/Pt [25], Pt/TiO ₂ /Pt [26], Pt/ZrO ₂ /Pt [27], Pt/HfO ₂ /Pt [28])	Mainly unipolar switching	Thermal dissolution model: Oxygen ions accumulated near the anode during the SET. The Joule heating activates O ²⁻ to combine with oxygen vacancies during the RESET and rupture of conductive filaments (CFs)
	Asymmetric structure: (e.g., Pt/TiO ₂ /TiN [29], TiN/ZnO/Pt [30], TiN/HfO ₂ /Pt [31], Pt/NiO/Cu [32], Cu/La ₂ O ₃ /Pt [33])	Mainly bipolar switching	Ionic migration model: Oxide interfacial layer formation between electrode and oxide acts as oxygen diffusion barrier.
	Structures	Mechanism	Advantages
<i>RRAM electrodes</i>			
Active and inert metal electrode	Active metal electrode: Ti, Al, Ag, Cu, Ni, TiN, TaN, ZrNx, ITO	ECM VCM Mainly bipolar switching	Active + inert electrodes in RRAM exhibit better endurance and longer retention time
	Inert metal electrode: Pt, Au, Ru, Pd	VCM Mainly unipolar switching	
	Novel electrodes: CNT [34, 35] n + Si [36] graphene [37]		
	Structures	Mechanism	Advantages
<i>RRAM switching layer</i>			
Doping	Dopants: Ti [20], Ge [21]	Forming-free: Dopants diffuse into oxide layer and oxygen vacancies are increased	Good data retention, potential multi-bit operation, highly scalable property and fast switching speed Better uniformity (temporal fluctuations) (cycle to cycle) and spatial fluctuations (device to device)
Bilayer	One layer with lower resistivity and one with higher resistivity E.g., TMO ₁ /TMO ₂ [24], Ni/GeO _x /HfO _x /TaN [38], MoO _x /GdO _x [39]	Resistive switching occurs at higher resistivity layer, and the lower resistivity layer can improve the resistance of the ON state	Low power consumption Lower RESET current Better uniformity (diffusion of metal ions from the lower resistivity to the higher resistivity layer. This stabilizes the CFs to reduce the randomness of resistive switching (RS))
Multi-layer	E.g., Pt/HfO _x /TiO _x /HfO _x /TiO _x /TiN [40], Pt/Ta ₂ O ₅ /TiO _x N _y /TiN/Ta ₅ /Pt [41]		Multi-level storage function

chip design. When compared with other types of synaptic device, RRAM possesses the advantages of high endurance that FLASH memories do not offer, lower programming energy and smaller cell size than

MRAM. In addition, the non-volatile property of RRAM outweighs the traditional DRAM which requires constant recharging [82–87]. Table 3 lists out some of the leading institutions and industries using

Table 2 Conduction mechanism analysis

Mechanism	Remarks
P-F emission (field-assisted thermal ionization)	Thermal excitation of electrons emits from traps into the conduction band of the dielectric. $\ln(1/V) - V^{1/2}$
SCLC (space charge limited current)	When electric field is high, the current is dominated by the charge carrier injected from the electrode. The current is only dependent on the mobility rather than the charge-carrier density. $I \sim V^2$
Conductive filament	Formation and rupture of conductive filaments $I \sim V$ Not proportional to electrode area
Electrode limited	Relevant to electrode (materials, area)
Schottky emission	Thermal excitation of electrons emits over the barrier into the conduction band of the dielectric.
Direct tunneling	Electron tunnels from cathode to anode directly Thin film thickness less than 3 nm
Fowler–Nordheim (F-N) tunneling	Electron tunnel from cathode to conduction band directly High voltage

**Figure 4** Interconnectivity in neuronal circuit.

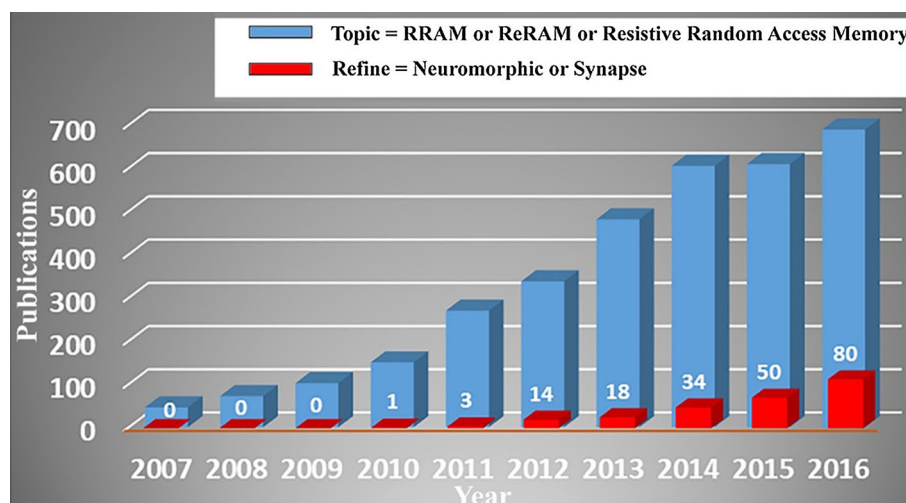
resistive memories as synaptic devices. Phase change memory (PCM) has shown great potential for neuromorphic computing in the Synapse project performed by leading researchers from IBM (TrueNorth) [88]. However, the greatest drawback is its high

power consumption. RRAM is touted as a possible solution and has been studied by Nanoelectronics and Nanotechnology Research Group from Stanford University and NanoST laboratory from National Chiao Tung University [89–92]. The results showed that RRAM possesses similar performance as that of PCRAM and consumes less power to operate artificial synapse [93–95]. Since then, RRAM has attracted massive attention from numerous academic institutions. From Fig. 5, the number of publications on RRAM has increased by 14 times in the last decade. Among these publications, the proportion of papers demonstrating neuromorphic computing applications has increased every year since 2010. In 2016, nearly 11% of papers presented RRAM's applications in the neuromorphic system. In general, most of studies on RRAM involve oxide materials. Oxide-based RRAM is attractive as the underlying metal insulator–metal structure is simple, compact and CMOS-compatible. Most importantly, the multi-level behavior needed to imitate adaptive synaptic changes

Table 3 Active players in the resistive memory of the synaptic devices (with to Ref. [97])

Project performer	Synaptic device	Device structure	Scale	Synaptic characteristics	Application
GIST	RRAM	TiO _x /HfO _x	Unit cell	Multi-level cell	Visual cortex
LETI	CBRAM	Ag/GeS ₂	8 × 8 1T1R	N/A	Human Cochlea and Retina
Stanford	PCM	GST	10 × 10 1T1R	Yes	Pattern recognition
POSTCH	RRAM	Pr _{1-x} Ca _x MnO ₃	11 k Array	Yes	Signal recognition
IBM	PCM	Ge ₂ Sb ₂ Te ₅	165 K Array	N/A	Pattern recognition

Figure 5 Publications per year from 2007 to 2016. The data is from the web of science site (www.webofknowledge.com). The search expression for blue bars is Topic = “RRAM” or “ReRAM” or “Resistive random-access memory”. The data for red bars are from subsequent refined with expressions Refine = “neuromorphic” or “synapse”.



has been found in many oxide-based RRAM devices. In addition, the energy per synaptic operation can be made as low as sub-pJ and the programming current can reach below 1 μ A [96]. In this review, we will focus on oxide-based RRAM memristor device as a synapse in the neuromorphic system.

The basis of learning and formation of memory are the modification of synapse connections as a result of accumulated experience, the feature known as synaptic plasticity. So far, synaptic plasticity had been widely studied their correlation in learning and memory [98–100]. Hebbin’s rule postulates that changes at synapses within the brain underlie learning and memory, which was proposed by Donald Hebb in 1949 [101]. It provides an algorithm to update weight of neuronal connection within neural network and a physiology-based model to reproduce the activity-dependent features of synaptic plasticity. In biology, excitatory and inhibitory postsynaptic potentials are transmitted between neurons through chemical and electrical messaging at synapses, driving the generation of the new action potentials (spikes). In RRAM-based neuromorphic approaches, the tunable resistive state of RRAM functions as synaptic weight in the neuromorphic system. The weight of synapse could be manipulated by changing the resistance states in RRAM devices, which follows the rule of spike-timing-dependent plasticity (STDP) [102–104]. STDP is a form of synaptic plasticity that adjusts the strength of the connection between the presynaptic neuron and postsynaptic neuron based on the relative timing of its output and input action potentials (electric pulses). In the implementation of STDP learning rule, it requires the adjustment of the

connection strength between pre- and postsynaptic neurons. The process of adjusting the strength of these interneural connections is known as synaptic operation. The number of synaptic operations of a neuromorphic chip is strongly dependent on the intrinsic properties of the materials used as the synapse. This will further determine the lifetime of the chip or its availability for learning process. The required cyclability (number of synaptic operations) of the chip will depend on the expected lifetime and operation frequency of the chip. If the chip is expected to be available for learning for about 10 years under 100 Hz of operation frequency, the chip must be able to execute at least 10^{10} synaptic operations. RRAM device has been reported to achieve endurance of as high as 10^{12} ; thus, it emerges as one of the strongest candidates for the development of highly reliable neuromorphic chip. The feasibility of RRAM device for synapse applications was first demonstrated in 2010 by Jo et al. [53]. According to the experimental RRAM STDP curve as shown in Fig. 6, conductance increases with the time delay when the presynaptic spikes precede the postsynaptic spikes ($t < 0$). Conversely, the decrease of conductance occurs when the postsynaptic spikes precede the presynaptic spikes ($t > 0$). In addition to the rise/fall of synaptic weight, the change of synaptic weight could also be either long term or short term. Upon changing the spike rate, Ohno [105] demonstrated both the long-term potentiation and short-term potentiation in Ag_2S -based RRAM device. Long-term potentiation refers to the persistent increase in synaptic strength which lasts minutes or more, while short-term potentiation refers to the transiently

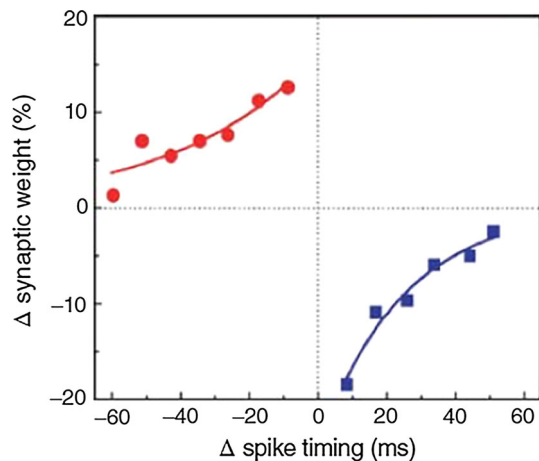
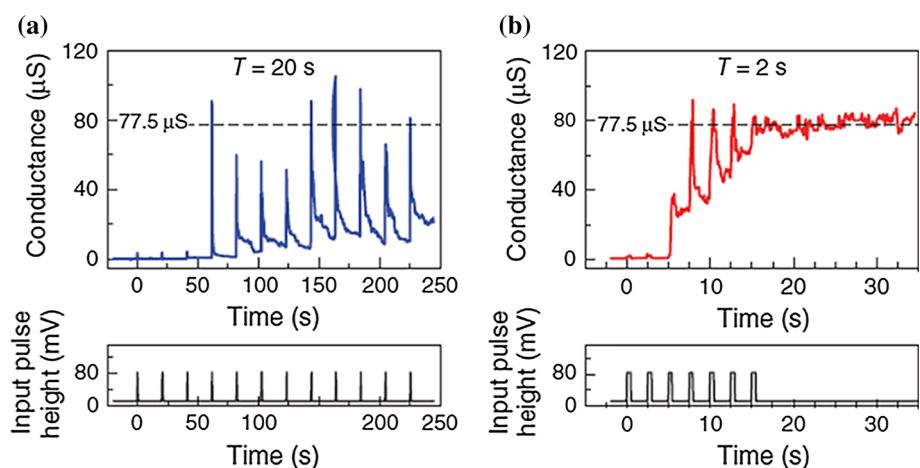


Figure 6 Experimental RRAM STDP curve. (Reprinted with permission from Ref. [53].).

enhancement of synaptic strength which acts on a timescale of tens of milliseconds to a few minutes. The results showed that the RRAM device exhibits short-term potentiation behavior with the applied pulse width at 0.5 s and interpulse delay at 20 s. As shown in Fig. 7, the potentiation of synaptic weights is volatile and no permanent change was observed after nine pulses. However, when the interpulse delay was reduced by 10 times, 6 pulses are sufficient to induce the stable potentiation and permanent high conduction state remains thereafter. In general, short-term and long-term plasticity could be mimicked in the single synapse by tuning the non-overlapping spike rate. This synaptic plasticity plays a crucial role in the way the brain implements learning and memory. Therefore, the rule of STDP (Fig. 7) is considered as one implementation of Hebbian learning.

Figure 7 **a** Short-term potentiation and **b** long-term potentiation of Ag_2S -based devices are induced for different spike rates. (Reprinted with permission from Ref [105].).



Oxide-based RRAM for neuromorphic computing application

Resistive switching behavior had been observed in many oxides, while most of them are transition metals. Some of them had been explored for neuromorphic application and summarized in the periodic table (Fig. 8). We selected PMCO, HfO_x , TaO_x , TiO_x , NiO_x , AlO_x , WO_x and other oxides for discussion in this article as their property and application for neuromorphic computing were well investigated. The growth techniques of RRAM oxide materials include traditional deposition methods, i.e., sputtering, atomic layer deposition (ALD) and pulsed laser deposition (PLD). In addition, novel methods for RRAM fabrication have also been reported. The traditional and novel methods for RRAM are listed in Table 4.

$\text{Pr}_{1-x}\text{Ca}_x\text{MnO}_3$ (PCMO)-based synaptic devices

$\text{Pr}_{1-x}\text{Ca}_x\text{MnO}_3$ (PCMO), as representative materials in perovskite transition metal oxide, has been widely employed as synaptic devices. Its resistive switching property was first discovered by Asamitsu et al. in late 1990 [135], which draws the attention from academies on complex transition metal oxide for RRAM device [136, 137]. The resistive switching of PCMO is based on metal–insulator (MIT) mechanism. MIT is a kind of valence change system, in which the electrons are injected into the insulator layer to conduct the current with an external applied field. In PCMO, the injected electronic charge distorts the superlattice structure and the mixed valence band behavior, a

The Periodic Table of the Elements

Corresponding Oxide has resistive switching behavior

Corresponding Oxide reported for use in neuromorphic computing

*Pr, Ca, MnO₂ was reproted for use in neuromorphic computing as compound

Figure 8 The periodic table of the elements highlighted with the host metal of RRAM materials and those had demonstrated for neuromorphic computing.

Table 4 Summary of RRAM device fabrication methods

RRAM device deposition methods

Traditional methods

Sputtering [106–113]
 Atomic layer deposition [23, 114–117]
 Pulsed laser deposition [118–122]
 Plasma enhanced chemical vapor deposition [123–125]
 Chemical synthesis particle + spin-coating [126–128]
 Sol-gel [129–132]

Novel methods

Shattering process using an atomic force microscopy (AFM) tip [133]
 Semi-auto screen printer [134]

process similar to ion doping processes. In addition, PCMO is LRS active materials in which the SET operation is associated with reverse biasing, while RESET operation occurs at forward biasing.

In 2012 and 2013, Park [138, 139] demonstrated 1-kbit cross-point array neuromorphic system based on PCMO synaptic devices. The HRS is achieved with the applied positive bias when the oxygen ions are attracted from the PCMO layer, thus forming a thicker oxide layer with Al electrode. The continuously increasing potentiation and depression behaviors were found when the identical spikes (pulses with same amplitude and width) were applied. In order to overcome the difficulty in achieving gradual long-term depression (LTD) caused by asymmetry potentiation/depression, a new programming scheme was developed [140]. The growth process control of PCMO device with TiN electrode was

studied to optimize the device electric performance [141]. Similarly, to avoid an abrupt LTD, a two-PCMO-memristor device model was proposed by Moon et al. [142]. In 2015, a circuit capable of realizing the learning process was designed and published [143]. In the same year, a high-density PCMO cross-point synapse array on the 8-inch wafer was fabricated. The proposed system had been demonstrated to recognize human thought patterns for three vowels [97]. In addition, vivo experiment had also been performed by the group. In the paper, the PCMO-based nanoscale analogy synapse device was successfully applied for neural fear-conditioning signal recognition on a live rat. In 2016, Moon, K from Hwang, H group reported PCMO-based interface switching device with 5-b MLC (32 levels) and improved data retention by using Mo electrodes for the neuromorphic system [144]. The Mo/PCMO

synapse array has experimentally confirmed with the realization of pattern recognition with high accuracy when working with NbO_2 oscillator neuron [145]. In 2017, a comprehensive study of PCMO-based synaptic devices had been reported in the book “Neuro-inspired Computing Using Resistive Synaptic Devices” [146]. The PCMO synapse performance is evaluated in 1-kb and 8-kb crossbar arrays. The interfacial-type PCMO RRAM was demonstrated with wide on/off ratio (10^4), extremely stable analogy resistance changes and low LRS current (1 μA) when device scales down to 150 nm. The influence of active electrode, nitrogen treatment and extra insulating layer (AlO_x) has also been investigated. The synaptic performance of PCMO is evaluated as shown in Fig. 9, where the potentiation and depression with identical and non-identical spikes are displayed. It is observed that the gradual switching of depression is only triggered by non-identical spikes, which is accounted for the high asymmetry ratio of potentiation/depression conduction.

So far, PCMO is one of the most matured RRAM materials for neuromorphic computing. It possesses various advantages: nanoscale device dimension (150 nm), multi-level states (5-b MLC), retention, uniformity ($\text{STD}(\sigma)/I_{\text{ave}} < 0.5$), on/off ratio (10^4) and high-density crossbar array structure (8 kb) [146]. However, further improvement of symmetric weight update is still required.

HfO_x-based synaptic devices

HfO_x materials have been widely used in RRAM materials because of its excellent CMOS compatibility. In industries, HfO_x has been employed as a high-k dielectric for the gate insulator of CMOS MOSFETs. In HfO_x-based RRAM device, TiN is usually applied as an electrode. TiN functions as an oxygen scavenger that depletes the O atom from HfO_x thin film and functions as an oxygen reservoir. It has been reported that HfO_x RRAM has a large on/off ratio ($> 10^2$), high endurance ($> 10^6$), retention, multi-bit storage and high-speed operation (< 10 ns) [142–146].

In 2011, a novel electronic synapse of HfO_x/AlO_x-based memory was presented by Yu et al. The structure showed attractive features such as reduced randomness of resistive switching, multi-level switching and the capability to modulate resistances based on the input pulse amplitudes, suggesting great potential to use in emerging neuromorphic computation system [147]. In 2012, the same group reported HfO_x-based RRAM device application in the neuromorphic visual system. At the system level, a neuromorphic visual system consisting of 1024 CMOS neuron circuits and 16,348 RRAM synaptic devices was fabricated. The system exhibited great performance in image orientation selectivity and high tolerance on temporal and spatial variability. After one year, Gao, B [148] and his coworkers developed a method to achieve gradual switching on the setting process in TiN/HfO_x/Pt devices. To avoid the generation of random and avalanching oxygen vacancies

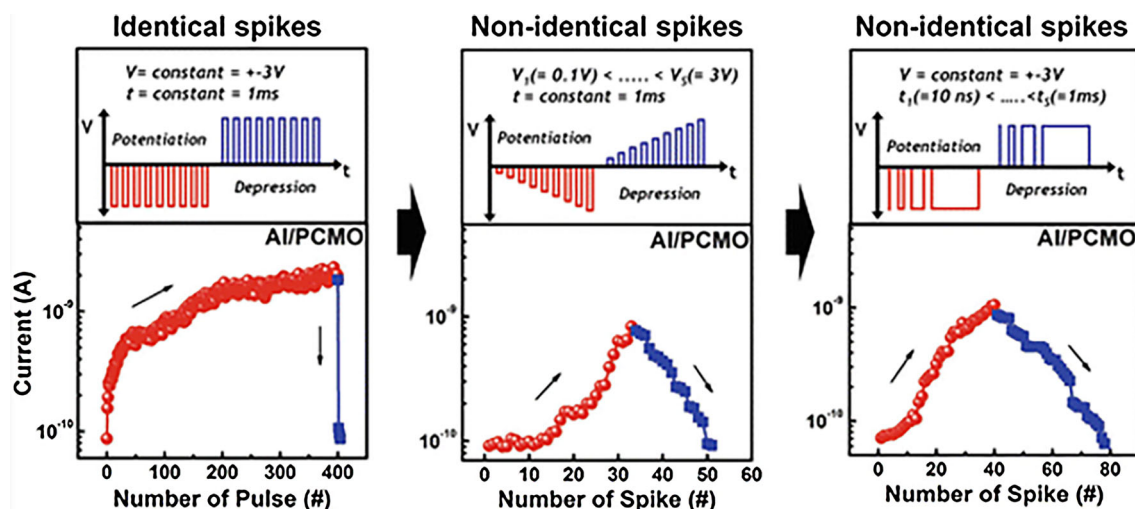


Figure 9 Potentiation and depression with various programming spikes: identical and non-identical spikes, reprinted with permission from Ref. [146].

(V_O) clusters, the team developed a methodology by doping trivalent elements into HfO_x layer. The local formation energy of V_O near the dopants was reduced, and V_O will distribute more uniformly in the conductive filament region. The fabricated $TiN/Gd:HfO_x/Pt$ devices have also shown controllable multilevel resistive switching behavior. Garbin [149] applied pulse-train schemes to a 3 bit per cell HfO_2 RRAM in 2014, causing the relative standard deviations of resistance levels to improve up to 80% compared to the single-pulse scheme. In the same year, Benoist, A integrated $TiN/HfO_2/Ti/TiN$ RRAM with an advanced 28-nm CMOS process [150]. A global overview of HfO_2 material performances was assessed on a statistical basis, and projection for larger array integration was discussed. The influence of Cu dopant was investigated by Tingting, G., and it was found that Cu-doped HfO_2 -based RRAM has improved resistive switching with multilevel storage [151]. The application of Mn-doped HfO_2 -based RRAM has demonstrated speech recognition by Mandal [3]. A comparison between a 20-nm times 20-nm sized synaptic memory device with that of a state-of-the-art VLSI SRAM synapse has exhibited 10^3 reduction in area and 10^6 times reduction in the power consumption per learning cycle. The colormap for speech recognition of words “Hello” and “Apple” is shown in Fig. 10.

In 2015, Wang presented a new artificial synapse scheme, consisting of a HfO_x RRAM memristive switches connected to 2 transistors responsible for gating communication and learning operations. STDP was achieved through appropriate shaping of the presynaptic and the postsynaptic spikes [152]. Covi, E demonstrated the $Al:HfO_2$ memristor device use for artificial synapse, emulating the potentiation

and depression processes using an easily implementable algorithm based on a train of identical pulses [153]. The impact of HfO_x -based synapses cycle to cycle (temporal) and device to device (spatial) variability on the convolutional neural network was investigated by Garbin, D [154]. The results showed that the convolutional neural network (CNN) architecture has a high tolerance to the variability. Lee et al. [155] proposed a silicon-based charge trap memory with $Al/HfO_2/Al_2O_3/Si_3N_4/Si$ structure. The device implemented both short-term plasticity and long-term potentiation in the synapse. The use of HfO_2 -based oxide-based resistive memory (OxRAM) devices operated in a binary mode to implement synapses in a CNN was also studied by Garbin, D in 2015 [149]. The proposed HfO_2 -based OxRAM technology offers good electrical properties including high endurance ($> 10^8$ cycles), fast speed (< 10 ns) and low energy (< 10 pJ). High accuracy (recognition rate $> 98\%$) was demonstrated for a complex visual pattern recognition application. Covi, E proposed $TiN/HfO_2/Ti/TiN$ memristor as artificial synapse for neuromorphic architectures. The collected STDP data were used to simulate a simple fully connected spiking neural network (SNN) for pattern recognition [156]. An innovative approach for real-time decoding of brain signals based on SNN was presented by Werner [157]. HfO_x -based RRAM devices were used to implement synapse and SNN, enabling the network for autonomous online spike sorting of measured biological signals. The system allows real-time learning and completely unsupervised operation. The real-time functionality, low power consumption (10 nW) and high recognition rate 90% of the system make it a great candidate for future healthcare applications. HfO_x -based synapses were used in SNN

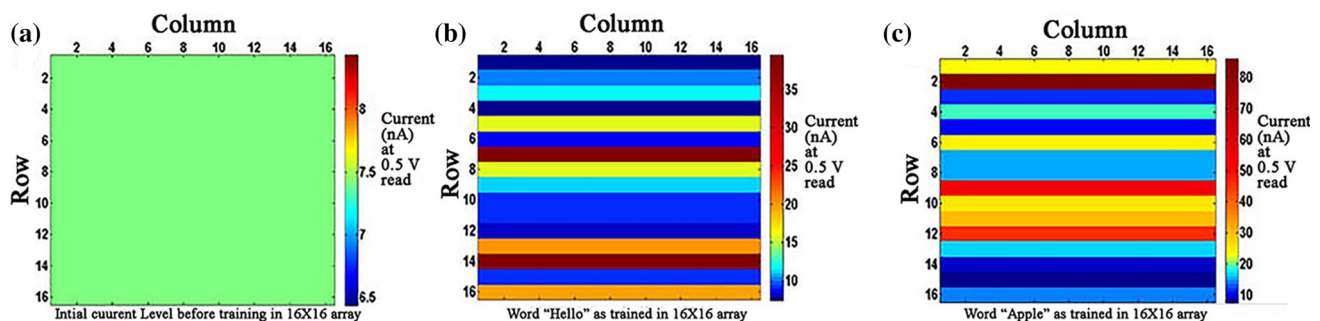


Figure 10 Synaptic weights. **a** Initial current level. **b** The conductance distribution when the word “Hello” trained on the crossbar array of synaptic devices. **c** The conductance distribution

when the word “Apple” trained on the crossbar array of synaptic devices. Reprinted with permission from Ref. [3].

for real-time unsupervised spike sorting of complex brain signals. Synaptic weights are modulated through the application of an online learning strategy inspired by biological STDP. According to the report by Wang et al. [158] this network had been tested by real spiking data from the *in vitro* preparation of the crayfish sensory-motor system. The proposed RRAM was validated by different sets of real biological spiking data without parameter tuning (e.g., the threshold level for spike detection). This artificial SNN is able to identify, learn, recognize and distinguish between different spike shapes in the input signal with a recognition rate about 90% without any supervision.

HfO_x is a superior RRAM material and the HfO_x-based synaptic device has demonstrated its application in neuromorphic computing, such as speech or pattern recognition. Gradual switching in the SET process can also be achieved through doping with trivalent metals, enabling the high-density applications.

TaO_x-based synaptic devices

TaO_x-based synaptic devices have also attracted much attention. TaO_x switching layer in an RRAM device usually consists of two phases: TaO₂ phase which is more conducting and Ta₂O₅ phase which is more insulating. The TaO_x RRAM has been reported to possess high endurance ($> 10^{12}$ cycles), and its application in neuromorphic computing has been well demonstrated. TaO_x could be engineered in synaptic device alone or combined with another oxide (TiO₂) layer in MIM structures. In this section, both single-layer TaO_x or multi-layer TaO_x-based synaptic devices will be discussed.

Wang reported a high-density 3D synaptic architecture based on Ta/TaO_x/TiO₂/Ti RRAM with ultralow sub-10 fJ energy per spike for neuromorphic computation in 2014 [159]. In addition, the analogue synaptic plasticity was simulated using the physical and compact models to facilitate future neuromorphic system designs. Thereafter, Wang et al. further investigated the synaptic device with the same structure experimentally and theoretically. The results showed that the device can be used for implementing concurrent inhibitory and excitatory synapses (Fig. 11). In addition, the devices also provided superior performances as compared to typical filamentary synaptic devices in reducing synaptic

conductance and state fluctuation [160]. In the report, a physics-based compact model was also proposed for facilitating circuit-level design. Based on previous studies, the group successfully applied 3D two-layered Ta/TaO_x/TiO₂/Ti cross-point synaptic array to implement highly anticipated hardware neural networks (HNN) in 2016 [161]. In total, more than 50 analogue synaptic weight states were controlled with minimal drifting during a cycling endurance test of 5000 training pulses. The team also proposed a new state-independent bipolar-pulse-training (BP) scheme, and this scheme significantly improved the nonlinearity of weight updates, consequently improving the training accuracy.

Similarly, Ta/TaO_x/TiO₂/Ti synaptic devices were also reported by Gao et al. [162]. The device is forming-free, and more than 200 levels of conductance states could be continuously tuned by identical programming pulses. In addition, the team proposed a novel fully parallel write and read scheme to accelerate the weight update and increase energy efficiency in the training process on the chip. In order to engineer the conduction modulation linearity, Wang et al. [163] proposed a new approach in 2016 where a diffusion limiting layer (SiO₂) is inserted at the TiN/TaO_x interface. The device exhibited higher switching linearity and lower power consumption that is desirable in neuromorphic computing hardware. Recently, Pt/Ta₂O_{5-x}/W devices fabricated using vertical 3D architectures were demonstrated by Wang [164]. The proposed memory can be recovered with a timescale when the electrical stimulation was removed. This recoverable process can emulate the synaptic plasticity including rapid decay and slow decay stages of forgetting in the memory loss process of the human brain. Yao, P and his coworkers reported face classification using electronic synapses of TaO_x/HfAl_yO_x-based RRAM devices [165]. This optimized device structure has shown bidirectional continuous weight modulation behavior. A neuromorphic network is developed using analogue 1024-cell-1T1R RRAM array, and its application in grayscale face classification has been experimentally demonstrated with parallel online training. The energy consumption in each iteration is $1000 \times (20 \times)$ smaller than the Intel Xeon Phi processor with off-chip memory.

From our literature review, bilayer stack TaO_x/(HfAl_yO_x, TiO_x) is intentionally designed as a synaptic device in the application of neuromorphic

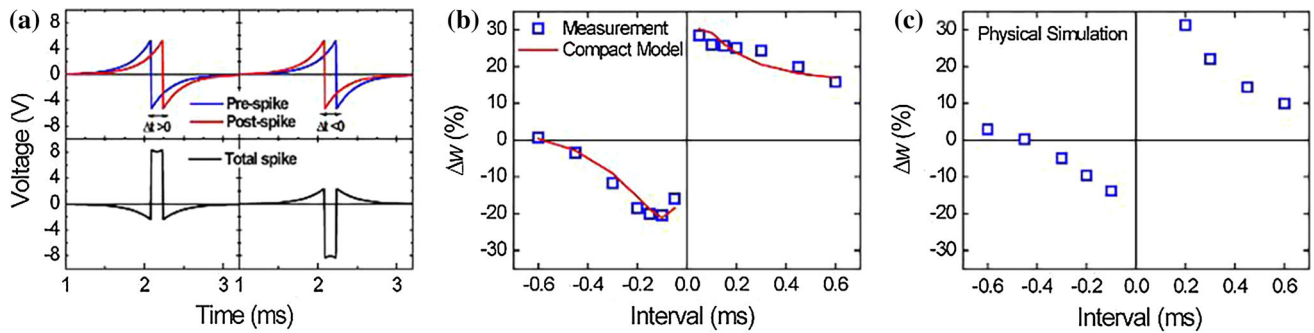


Figure 11 Spike-timing-dependent plasticity (STDP). **a** Biomorpho-action-potential-like waveforms used for STDP measurement and simulation. The prespike pulse was applied to the Ta electrode, and the postspike pulse was applied to the Ti electrode. The potentiation and depression actions were controlled by the relative timing of the pre- and postspikes (Δt). The synapse weight change

(Δw) was positive for $\Delta t > 0$, whereas it was negative for $\Delta t < 0$. **b** The measured Δw as a function of Δt for the Ta/TaO_x/TiO₂/Ti device. The red line shows the fitting result of using the physics-based compact model. **c** Simulation results for STDP, obtained by considering oxygen ion migration and the homogeneous barrier modulation model. Reprinted with permission from Ref [160].

computing. The possibility of engineering the TaO_x-based RRAM array in 3D structure enables the emulation of the high-density synaptic network in the real biological system.

TiO_x-based synaptic devices

TiO_x is one of earliest materials found to exhibit resistive switching property. In the previous section, TaO_x/TiO_x bilayer structure RRAM for synaptic application has been introduced. Here, we will discuss other multi- and single-layer TiO_x-based synaptic devices.

In 2011, the synaptic performance of TiO was explored by Seo [46], demonstrating that TiO_x and TiO_y bilayer RRAMs exhibit analogue property where the multilevel conductance states were caused by the movement of oxygen ions between the two TiO_x phase. The STDP was demonstrated by applying 100 successive identical pulses for potentiation or depression of synaptic weight. A year later, Yu developed TiO_x/HfO_x/TiO_x/HfO_x multi-layer RRAM stacks exhibiting more gradual and smooth RESET switching for synaptic devices [166]. Short pulses (10 ns) with an identical pulse amplitude enabled a sub-pJ energy per spike with potentially simple neuron circuits. Berdan emulated short-term synaptic dynamics with TiO₂ memristive devices in 2015 [167]. The meta-stable memory transitions in TiO₂ RRAM devices have proved to be the key feature to capture short-term synaptic dynamics. In addition to Seo, K works on bilayer TiO_x/TiO_y RRAM structure in 2011 [46], Bousoulas, P studied

amorphous–crystalline interfaces in TiO_x/TiO_y RRAM structures for enhanced resistive switching and synaptic properties [168]. A physical model was also proposed to divulge the crucial role of temperature, electric potential and oxygen vacancy density on the switching effect. An 8 × 8 array of the neuron on a standard 6 M 180 nm of CMOS process as part of a larger multi-purpose neuromorphic chip was demonstrated by Mostafa, H as shown in Fig. 12 [169]. The proposed CMOS-memristor system comprises of CMOS neurons interconnected through TiO_x memristors and spike-based learning circuits which modulate the conductance of the memristive synapse elements according to a spike-based perceptron plasticity rule. In 2016, Park developed a Mo/TiO_x-based interface RRAM and proposed a hybrid pulse mode for the synaptic application [170]. The TiO_x-based device is capable of producing 64-level conductance states, and the proposed hybrid pulse mode improves the symmetry of conductance change under both potentiation and depression conditions. The neural network simulation has shown that the hybrid pulse mode can enhance the pattern recognition accuracy of the proposed TiO_x stack.

Although extensive investigations have been carried out on switching mechanism of TiO_x RRAM, the demonstration of TiO_x RRAM for neuromorphic computing is still limited when compared to the above-mentioned PCMO, HfO_x, TaO_x materials. In most situations, TiO_x is used in conjunction with bilayer or multi-layer synaptic device to optimize the performance.

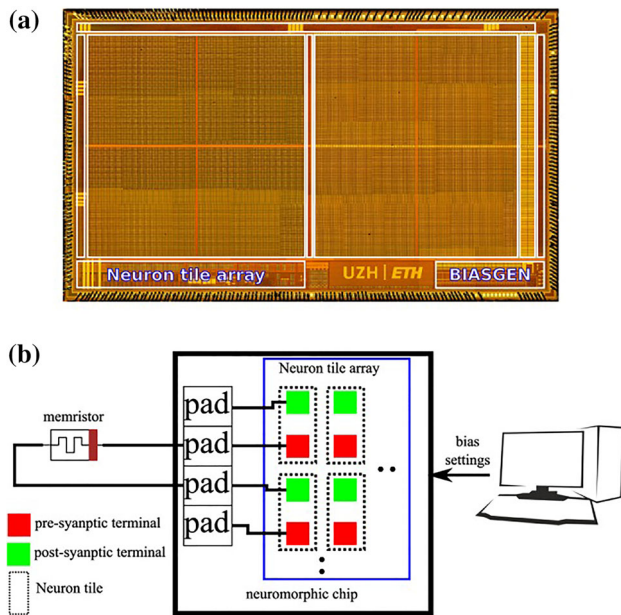


Figure 12 **a** Micrograph of the multi-purpose neuromorphic chip die showing the neuron tile array and the bias generator. **b** Illustration of the hardware setup used to obtain measurements in this paper. The presynaptic terminal of one neuron tile is connected to the postsynaptic terminal of another neuron tile through an off-chip TiO_{2-x} memristor. A PC controls the digital settings of the on-chip bias generator. Reprinted with permission from Ref. [169].

NiO_x-based synaptic devices

Similar to TiO_x , NiO_x is one of earliest materials found to exhibit restive switching behavior. Although NiO_x -based RRAM devices have been reported with high endurance (10^6) and retention, its application for neuromorphic computing is restricted due to the poor uniformity. A bipolar NiO_x RRAM, two NMOS FET and a capacitor were used to fabricate analogue synaptic devices (circuits) by Akoh [171]. This device also has the ability to update the synaptic conductance according to the difference of pre- and post-neuron spike timing. In 2013, Hu emulated the paired-pulse facilitation of biological synapse with NiO_x -based memristor, thus creating a form of short-term synaptic plasticity [172]. It was also observed that the current in NiO_x -based memristor had increased after a second electrical pulse. Additionally, the magnitude of the facilitation decreases with the pulse interval and it increases with the pulse magnitude or pulse width. Hu, S.G and his coworkers realized the well-known Pavlov's dog model with NiO_x -based memristor [173]. The long-term

potentiation behavior was emulated by the increase in memristor's conductance when applying the electrical pulses. In addition, spontaneous conductance decaying toward its initial state resembles the synaptic long-term potentiation. Finally, an artificial neural network was constructed to realize the Pavlov's dog model (Fig. 13).

WO_x-based synaptic devices

Tungsten oxides (WO_x) are another great candidate for memristive devices as a bio-inspired artificial synapse. The advantage of WO_x -based synaptic devices includes accredited endurance (10^5 cycles), CMOS compatibility and memorization and learning functions. Here, we highlight the implementation of this material in neuromorphic computing as a biological artificial synapse.

Synaptic behaviors and modeling of a WO_x memristive device were reported by Chang [174]. The Pd/ WO_x /W-structured memristor device shows reliable synaptic operations with good endurance. The synaptic devices can endure at least 10^5 potentiation/depression pulses without degradation. Furthermore, the memristor behavior was explained by a novel model that takes both drift and diffusion effects into consideration. Figure 14 presents the retention loss

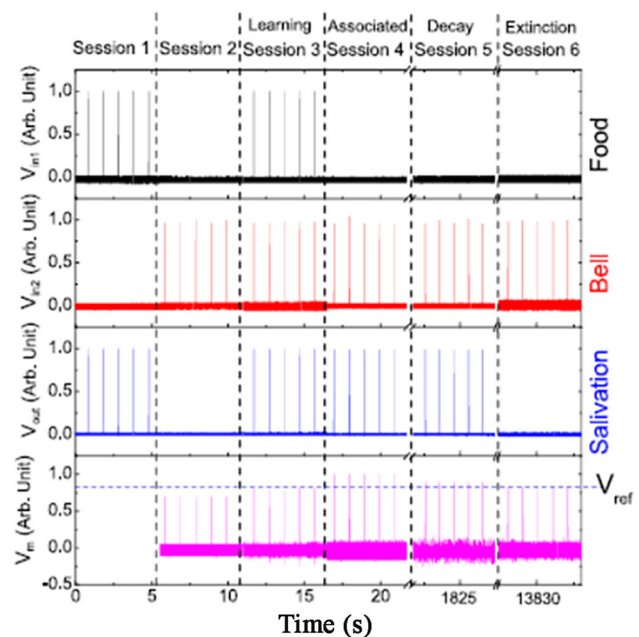
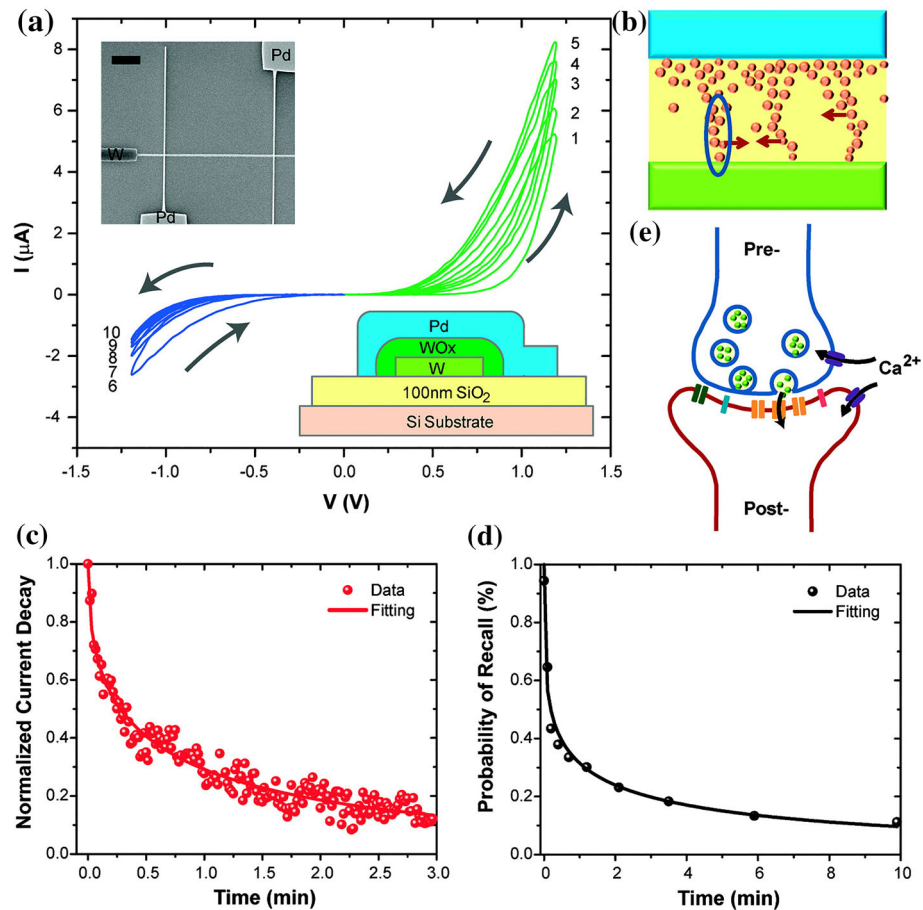


Figure 13 Learning and forgetting of association in Pavlov's dog experiment realized with the three-neuron neural network. Reprinted with permission from Ref [173].

Figure 14 **a** DC I - V curves of a memristor studied. **b** Schematic illustration of oxygen vacancy diffusion. **c** Retention loss curve of Pd/ WO_x /W-based memristor. **d** Forgetting memory curve. **e** Schematic illustration of synaptic plasticity modulation. Reprinted with permission from Ref [47].



curve and forgetting memory in a human memory curve from a paper by Chang [47]. It was found that the trend of retention curve of a Pd/ WO_x /W-based memristor and human forgetting curves greatly correspond to each other and both can be fitted using a stretched-exponential function (SEF). In addition, the temperature and humidity impact on the performance of a WO_x memristive device was studied by Meng et al. and Yong et al., respectively [175, 176]. According to the simulation results, the synaptic weight of WO_x memristive device decays faster as the temperature increases due to higher oxygen vacancies diffusion. In addition to the temperature, the memristive effects of tungsten oxide are also highly humidity dependent. The adsorbed moisture on the surface of WO_3 has resulted in decreasing conductances as the H cation induces an increase in barrier heights.

Among other transition metal oxides, WO_x is a great candidate material for synaptic devices application. For further exploring its application in

neuromorphic computing, enhancement of synaptic operation time (endurance) is of importance.

AlO_x -based synaptic devices

The resistive switching behavior in AlO_x was first reported by Hickmott [1]. AlO_x is of interest in RRAM materials due to its large band gap (~ 9 eV) and low RESET current ($\sim \mu\text{A}$). For neuromorphic application, AlO_x can also be used alone or stacked with other RRAM materials to improve the uniformity of the synaptic device characteristics.

AlO_x -based resistive switching device was investigated by Wu [49] to serve as a potential electronic synapse device. The $0.48 \mu\text{m} \times 0.48 \mu\text{m}$ Ti/ AlO_x /TiN memory stack exhibited multi-level resistance states when varying the compliance current levels or the applied voltage amplitudes during pulse cycling. In his work, around 1% resistance change per pulse cycling was obtained and an STDP realization scheme was proposed to implement the synaptic device in large-scale hardware neuromorphic

computing system. In 2015, Prezioso experimentally characterized $\text{Al}_2\text{O}_3/\text{TiO}_2$ -based memristors to modulate the impact of conductance-dependent conductance change on short-term dependent plasticity [177]. The utilization of $\text{TiO}_x/\text{Al}_2\text{O}_3$ -based memristor devices to emulate biological synaptic behavior for building brain-inspired computers was explored by Banerjee, W. et al. in 2017 [178]. The proposed $\text{TiO}_2/\text{Al}_2\text{O}_3$ synaptic device has the capability to switch from short-term memory (STM) to long-term memory (LTM) by introducing a mezzanine state at medium memory (MM) as shown in Fig. 15. In the paper, the dependence of retention (decay time) on conductance levels (memory level) was investigated and an experimentally proven psychological model is presented.

Till now, the potential validity of Al_2O_3 materials for electronic synapses had been experimentally demonstrated. Nevertheless, the reports of sophisticated Al_2O_3 -based memristor as an artificial synapse are limited. Therefore, further verification with different structures is required.

Other oxide-based synaptic devices

Besides the materials discussed above, a variety of other materials has been studied to implement neural network as a synaptic device. FeO_x -based RRAM with mixed-analogue-digital behavior was investigated by Deng et al. [12] for a recurrent neural network using the recursive least-squares algorithm.

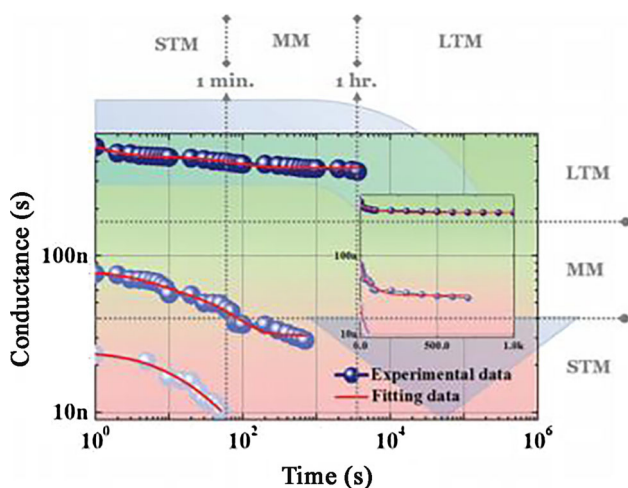


Figure 15 Memory effects in the $\text{TiO}_x/\text{Al}_2\text{O}_3$ -based electronic synaptic junction. The figure shows the conductance failure with time. A linear scale plot is shown in the inset of the figure. Reprinted with permission from Ref [178].

Wang, C.H et al. reported the synaptic behavior of a FeO_x -based RRAM in which the dependence of compliance current on conductance stability was studied [179]. The application of CeO_2 -based RRAM for highly energy efficient neuromorphic systems was demonstrated by Kim et al. The $\text{Pt}/\text{CeO}_2/\text{Pt}$ device exhibited the polarity-dependent and asymmetric diode-type resistive switching [180], and the detailed switching characteristics of artificial synapses (potentiation and depression) were discussed. A GdO_x and Cu-doped MoO_x stack with a platinum top and bottom electrodes were reported by Choi [45]. The weighted sum operation was carried out on electrically modifiable synapse array circuit based on the proposed stacks [181]. The biological synaptic behavior was demonstrated by Chang through integrating SiO_x -based RRAM with Si diodes. The proposed one-diode-one-resistor (1D-1R) architecture not only avoids sneak-path issues and lowers standby power consumption, but also helps to realize STDP behaviors [182]. VO_x is a well-known Mott material which experience sharp and first-order metal-to-insulator transition (MIT) at the around 68°C [183]. The application of VO_x as RRAM materials had been explored by Drisoll et al. [184] through sol-gel technique. Nevertheless, most researches on VO_x so far focus on its use for select device which can be integrated with RRAM device to mitigate sneak-path current. The $\text{Pt}/\text{VO}_2/\text{Pt}$ selector has been integrated with NiO unipolar RRAM by Lee et al. [185] in 2007 and $\text{ZrO}_x/\text{HfO}_x$ bipolar RRAM by Son et al. [186] in 2011. In 2016, 1S1R configuration of $\text{W}/\text{VO}_2/\text{Pt}$ selection device and $\text{Ti}/\text{HfO}_2/\text{Pt}$ RRAM was demonstrated by Kailiang et al. [187]. However, thermal instability is a major challenge for VO_2 for practical applications [14].

Summary and outlook

In conclusion, we have outlined an overview of oxide-based RRAM materials for the applications in neuromorphic computing. Table 5 summarizes the RRAM materials and their parameters discussed in this review. Our work suggested that the neuromorphic approach with oxide-based RRAM devices is promising. So far, PCMO, HfO_x , TaO_x , TiO_x , NiO_x , WO_x , AlO_x , etc., materials have been demonstrated for their application in oxide-based synaptic devices. However, challenges still remain in specific materials,

Table 5 Summary of RRAM materials in this review

Material	On/ off ratio	Endurance	Energy consumption	Multi- levels	Switching time	Neuromorphic computing applications
PCMO [97, 136–146]	10^3	10^5	6 pJ	32	8 ns	Pattern recognition Human thought patterns Neural fear-conditioning signal recognition
HfO _x [147–151]	10^2	10^6	10 pJ	8	10 ns	Speech recognition Visual pattern recognition
TaO _x [158–164]	10X	10^{12}	10 fJ	200	105 ps	Grayscale face classification
TiO _x [46, 166–170, 189]	10^5	2 X 10^6	sub-pJ	64	5 ns	Pattern recognition in simulation
NiO _x [171–173, 185, 190, 191]	10^6	10^6	–	5	20 ns	Learning and forgetting of association in Pavlov's dog experiment realized.
WO _x [47, 174–176, 192]	10^3	10^5	–	8	50 ns	Human forgetting curves
AlO _x [49, 193–195]	10^6	10^5	1.5 pJ	10	10 ns	Spike-timing-dependent plasticity realization scheme.
FeO _x [179, 196, 197]	10^2	6×10^4	–	6	10 ns	Long-term potentiation and long-term depression demonstration
CeO ₂ [180, 198]	10^5	10^4	–	8	200 ns	Synaptic potentiation and depression characteristics demonstration
MoO _x [39, 199]	$10 \times$	10^6	–	8	1 ms	Weighted sum operation carried out on MoO _x - based electrically modifiable synapse array circuit
SiO _x [182, 200–202]	10^7	10^6	–	4	100 ps	Long-term potentiation and long-term depression demonstration

e.g., poor symmetric weight update for PCMO_x-based synaptic devices and low endurance for WO_x-based synaptic devices. In general, there are two main key challenges for overall oxide-based RRAM materials. Although the inherent fault tolerance of neural network models is able to mitigate the impact of device variation to some extent, the improvement of spatial variation and temporal variation turns out to be one of the greatest challenges on a long-term basis. In addition, improvement of reliability characteristics of the RRAM synaptic devices is another key challenge which is not well studied.

In future material research of neuromorphic computing applications, the study of novel tunable materials with enhanced properties for neuromorphic devices is one of the most exciting components. One such example is MgO, which has a large band gap that ensures sufficient band offsets, a high dielectric constant that might potentially reduce leakage current (thus improving device scalability), a

high thermal conductivity which minimizes self-heating during operations and a large breakdown field that might prevent the device from irreversible damages. In addition, MgO-based magnetic tunnel junction memristors had been proposed for implementation of synapses [188]. Further research is required to expand the database of materials for synaptic devices and neuromorphic applications.

Finally, to implement oxide-based RRAM devices as a synapse in neuromorphic systems, it is imperative to have an in-depth understanding of the meta-plasticity mechanism and internal states of these memristive devices. The underlying mechanisms governing RRAM devices will inevitably be discovered from investigations of conduction and resistive switching mechanism via results from either experiment or simulation. Further research work incorporating interactions between materials, device levels, circuit designs and computing processes will

certainly speed up the realization of oxide-based RRAM synapses for neuromorphic systems.

Acknowledgements

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